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Expansion of harmful brown tides caused by the pelagophyte, *Aureoumbra lagunensis* DeYoe et Stockwell, to the US east coast

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ABSTRACT

Brown tides caused by the pelagophyte *Aureoumbra lagunensis* DeYoe et Stockwell have formed ecosystem disruptive algal blooms in shallow lagoons of Texas (TX), USA, for more than two decades but have never been reported elsewhere. During the summer of 2012, a dense brown tide occurred in the Mosquito Lagoon and northern Indian River Lagoon along the east coast of Florida (FL), USA. While chlorophyll *a* levels in this system have averaged 5 $\mu g\ L^{-1}$ during the past two decades, concentrations during this brown tide reached ~200 $\mu g\ L^{-1}$. Concurrently, levels of nitrate were significantly lower than average and levels of dissolved organic nitrogen were significantly higher than average ($p < 0.001$ for both). Sequences of the 18S rRNA gene of the bloom community and of single cell isolates were identical to those of *Aureoumbra lagunensis* DeYoe et Stockwell from TX. The *A. lagunensis* brown tide in FL bloomed to densities exceeding 10^6 cells mL^{-1} (quantified with a species-specific immuno-label) from July through September, began to dissipate in October, but maintained densities exceeding 10^5 cells mL^{-1} in some regions through December of 2012. The decline of the bloom was associated with near-hypoxic conditions and more than 30 fish kills reported within the Mosquito Lagoon in September 2012, a number far exceeding all prior monthly reports in this system dating to 1996. Wild northern quahog populations (a.k.a. hard clam, *Mercenaria mercenaria*) suffered mass die offs during the brown tide and eastern oysters (*Crassostrea virginica*) that settled during 2012 were significantly smaller than prior years. Clearance rates of hard clams and eastern oyster were significantly reduced in the presence of Mosquito Lagoon bloom water and *A. lagunensis* monocultures isolated from the Mosquito Lagoon at densities of ~10^6 cells L^{-1}. The expansion of harmful brown tides caused by *A. lagunensis* to these estuaries represents a new threat to the US southeast coast.

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1. Introduction

Harmful algal blooms (HABs) have become a common feature within coastal water bodies around the globe. These events can disrupt ecosystems and economies, as well as negatively impact human health (Van Dolah, 2000; Landsberg, 2002; Kirkpatrick et al., 2004). During the past two decades, the geographic distribution of several HABs has expanded, a trend that may be driven by climate change, ballast water transport, eutrophication, and improved detection capabilities (Smayda, 2002; Sellner et al., 2003; O'Neil et al., 2012; Anderson et al., 2012). One notable case of HAB expansion has been the harmful pelagophyte, *Aureococcus anophagefferens*. While brown tides caused by this species were first documented in the northeast US in 1985, they were subsequently detected in the mid-Atlantic US and South Africa during the 1990s (Trice et al., 2004; Probyn et al., 2001) and more recently in the Bohai Sea in northeast China (Zhang et al., 2012).

Another pelagophyte that forms harmful brown tides in the US is *Aureoumbra lagunensis* DeYoe et Stockwell. The first reported *A. lagunensis* brown tide occurred continuously from 1990 to 1997 in shallow, hypersaline lagoons in and around the Laguna Madre and Baffin Bay, TX, USA, where it has bloomed intermittently since (Buskey et al., 1997, 2001; DeYoe et al., 2007). These blooms were associated with a loss of seagrass and benthic invertebrates within affected ecosystems (Montagna et al., 1993; Onuf, 1996; Ward et al., 2000). *A. lagunensis* has a broad salinity tolerance, is known to inhibit grazing by micro- and meso-zooplankton, and is capable of growing rapidly in the presence of low levels of inorganic nutrients

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(Gobler and Sunda, 2012). While blooms of *A. lagunensis* have never been reported outside of Laguna Madre and Baffin Bay, low concentrations of *A. lagunensis* cells have been found in coastal waters across Texas, Mexico, and the west coast of Florida (Villareal et al., 2004).

Mosquito Lagoon and Indian River Lagoon are part of the Indian River Lagoon system and are connected to each other through Haulover Canal (part of the Atlantic Intracoastal Waterway). The Mosquito Lagoon is noted for its red drum fishery, supporting intertidal reefs of eastern oyster *Crassostrea virginica*, and a population of bottleneck dolphins (Johnson and Funicelli, 1991; Grizzle et al., 2002; Mazzoil et al., 2008). It has historically been characterized as an oligotrophic estuary as it generally harbors low levels of algal biomass (Phlips et al., 2002; Badylak and Phlips, 2004). During the summer of 2012, however, a dense and damaging phytoplankton bloom emerged across the Mosquito Lagoon and the northern Indian River Lagoon, FL, USA.

The goal of this study was to characterize a dense brown tide in the Mosquito and northern Indian River Lagoons, FL, USA, in 2012. We identified the alga responsible for the bloom as *Aureoumbra lagunensis* using genetic sequencing and transmission electron microscopy, and characterized the temporal and spatial extent of the bloom within the system using a species-specific antibody. We examined historical water quality and meteorological conditions to understand the conditions that may have contributed toward the formation of this bloom. In addition, we examined the ability of juvenile northern quahogs (*Mercenaria mercenaria*) and eastern oysters to filter feed in the presence of bloom water and mono-specific cultures of *A. lagunensis* isolated from the Mosquito Lagoon. Finally, we compiled information regarding fish kills, oyster recruitment, and water quality during the brown tide in the Mosquito and Indian River Lagoons. This bloom occurrence permitted hypotheses developed regarding brown tides in other regions (NY, TX) to be tested in a novel, ecosystem setting.

2. Methods

2.1. Study sites

This study investigated a novel brown tide event within the Indian River and Mosquito Lagoon systems that occurred during the second half of 2012. The Indian River and Mosquito Lagoons are part of Florida's shallow (~2 m deep) east coast barrier island estuary system (Badylak and Phlips, 2004; Phlips et al., 2010). This study focused on the northern part of the Indian River Lagoon system, near the city of Titusville, and the entire Mosquito Lagoon, to which it is connected through the Haulover Canal (Table 1 and Fig. 1). Both systems have long hydraulic residence times (range: 200–700+ days) primarily due to their distance from the nearest inlets to the Atlantic Ocean, Sebastian Inlet to the south and Ponce de Leon Inlet to the north, and relatively weak tidal forcing (Sheng et al., 1990; Smith, 1993; Steward et al., 2005). In addition, the watershed contributing to the northern Indian River and Mosquito Lagoons is relatively small, consisting primarily of undeveloped park land and conservation areas (i.e. Canaveral National Seashore, Turnbull Hammock Conservation Area) and is subject to only 30%

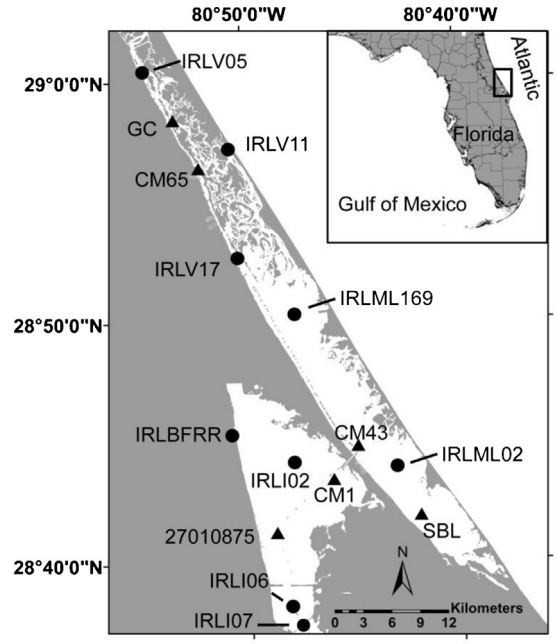

Fig. 1. Map showing the northern Indian River and Mosquito Lagoon systems on Florida's central east coast. Solid circles denote stations which are sampled routinely by the St. Johns River Water Management District while the triangles denote additional stations sampled as part of this study.

Table 1
Station location, physical and chemical parameters during a September 2012 survey of the Florida brown tide bloom. Sal, Temp, DO and Secchi stand for salinity, temperature, dissolved oxygen and Secchi disk, respectively. DO % denotes the % oxygen saturation. Stations at which water was collected for clearance rate experiments are marked with an asterisk.

Station	Date	Coordinates		Sal	Temp (°C)	Secchi (m)	DO (mg L^{-1})	DO (%)
		Longitude	Latitude					
IRLV05	09/24/12	29°00′26.9″	−80°54′36.5″	35.2	27.1	0.4	4.46	69.6
GC	09/24/12	28°56′24.1″	−80°52′04.8″	36.4	27.9	0.7	5.14	82.3
CM65	09/24/12	28°52′39.3″	−80°50′19.8″	37.1	28.2	0.6	6.07	97.7
IRLV11	09/24/12	28°50′16.4″	−80°47′42.8″	37.1	27.9	0.7	5.80	92.7
IRLV17	09/24/12	28°44′47.9″	−80°44′51.4″	36.4	27.6	0.5	5.07	80.1
IRLML169	09/24/12	28°43′54.5″	−80°43′02.4″	38.0	27.9	0.4	3.66	58.8
CM43	09/24/12	28°41′52.8″	−80°41′58.0″	39.2	28.5	0.3	5.09	83.6
IRLML02*	09/24/12	28°55′42.4″	−80°49′36.5″	38.0	28.3	0.4	6.71	108.9
SBL*	09/24/12	28°58′24.7″	−80°53′14.0″	36.6	28.2	0.7	5.97	95.8
IRLBFFR	09/25/12	28°43′24.5″	−80°46′01.9″	36.6	26.1	0.6	4.48	68.8
IRLI02	09/25/12	28°44′07.9″	−80°47′52.6″	34.7	26.4	2.0	3.17	48.5
CM1	09/25/12	28°46′18.1″	−80°50′17.4″	33.3	26.2	0.6	3.70	55.8
27010875	09/25/12	28°41′13.8″	−80°48′45.6″	33.2	26.7	1.6	4.77	72.8
IRLI06	09/25/12	28°38′09.9″	−80°48′08.0″	31.2	26.3	1.6	5.02	75.0
IRLML169	09/26/12	28°50′09.4″	−80°47′48.9″	37.0	26.8	0.4	5.33	83.4
CM43	09/26/12	28°44′48.6″	−80°44′53.8″	37.0	26.7	0.3	4.48	69.9
IRLML02*	09/26/12	28°43′54.6″	−80°43′02.5″	38.1	26.7	0.4	4.09	64.3
SBL	09/26/12	28°42′00.1″	−80°42′02.2″	38.5	27.4	0.7	4.00	64.1

of the mean annual runoff, compared to the central Indian River Lagoon (Phlips et al., 2010). Water samples were collected from July to December of 2012 from fifteen stations for this study (Fig. 1); eight of which are surveyed routinely as part of the St. Johns River Water Management District's (SJRWMD) water quality monitoring program.

2.2. Culture establishment, DNA analysis, and morphological observations

Culture isolates were obtained by serially diluting bloom water collected by the SJRWMD during August 2012 from the southern Mosquito Lagoon (station IRLML02; Fig. 1) into polystyrene cell culture plates containing GSe culture medium (Doblin et al., 1999) and an antibiotic–antimycotic solution (Mediatech, Inc., Hemdon, VA) added at a 1% concentration (final concentrations = 100 I.U. mL^{-1} penicillin, 100 $\mu g\ mL^{-1}$ streptomycin, and 0.25 $\mu g\ mL^{-1}$ amphotericin B) to ensure cultures remained free of bacteria and fungi. Cells were cultured in sterile GSe supplemented with the same antibiotic–antimycotic solution at 21 °C in an incubator with a 12:12 h light:dark cycle, illuminated by a bank of fluorescent lights that provided a light intensity of ~100 μEin quanta $m^{-2}\ s^{-1}$ to cultures.

For genetic analyses of culture isolates, 50 mL of cultured cells were pelleted by centrifugation at 5000 RPM for 2 min. The majority of the media was removed by aspiration, leaving behind ~150 μL with the cell pellet. The cell pellet was frozen at −20 °C overnight, then at −80 °C for 48 h, transferred to a 1.5 mL Eppendorf tube, and spun at 10,000 RPM for 3 min. The cell pellet was then resuspended and serially diluted (1:10) in LoTE buffer. Culture aliquots (1 μL) of the diluted series were used as PCR template (see below). For field samples, 50 mL of bloom water was collected on a 1 μm polycarbonate filter, placed in a 1.5 mL Eppendorf tube, and frozen at −80 °C. LoTE (500 μL; Tris–EDTA buffer) was added to the tube which was then vortexed and mixed via vigorous pipetting. Resuspended cells (50 μL) were removed and serially diluted (1:10) in LoTE. Aliquots (1 μL) of diluted field sample series were used as PCR template.

PCR reactions contained 1 μL of cellular resuspension, 12.5 μL 2× GoTaq Green (Promega), 1 μM of 18s rDNA forward primer (Euk A: 5-AAC CTG GTT GAT CCT GCC AGT-3), 1 μM of 18s rDNA reverse primer (329R: 5-TGATCCTTCYGCAGGTTCAC-3) and distilled, deionized water in a total volume of 50 μL. Reactions were performed with the following cycling parameters: 94 °C for 5 min, followed by 31 cycles of 94 °C for 45 s, 55 °C for 60 s and 72 °C for 3 min, followed by 72 °C final extension for 5 min. Sequencing was performed directly on the unmodified PCR product using 50 ng of PCR product and 3.2 pM of primer on a ABI3730 Genetic Analyzer using BigDye Terminator® Cycle sequencing kit (Applied Biosystems) at the Stony Brook University DNA Sequencing Facility. Sequences were analyzed with Geneious version *5.6.6* created by Biomatters (http://www.geneious.com/).

To preserve field samples for electron microscopy, equal volumes of buffered glutaraldehyde fixative and algal suspension from cultures were mixed to produce a final solution of 2% glutaraldehyde in 0.05 M cacodylate buffered seawater (pH = 8.0) at 5 °C. The fixed cells were gently sedimented to form a pellet, the supernatant was decanted, and 5 mL of 4% osmium tetroxide solution in the same cacodylate buffer was added to the pellet as a post stain and fixative. After 1 h at 5 °C, the osmium-fixed cells were sedimented, washed by addition of cacodylate buffer and again sedimented to form a pellet and enrobed in 0.8% agar matrix. Small segments of the agar-embedded pellet (~3 mm in size) were washed in deionized water, dehydrated in a graded acetone/aqueous series, embedded in TAAB epoxy resin (Energy Beam Sciences, Granby, CT) and polymerized in BEEM capsules at 70 °C for 18 h. Ultrathin sections, collected on uncoated copper grids, were obtained with a Porter-Blum MT-2 ultramicrotome fitted with a diamond knife, post-stained with Reynolds's lead citrate, and observed with a Philips 201 transmission electron microscope (TEM).

2.3. Analysis of water quality data sets and ecosystem impacts

To assess environmental changes that occurred before, during, and after the brown tide event of 2012, the long term water quality data set (1996–2012) generated by SJRWMD was assessed at the stations indicated in Fig. 1. Data specifically examined included monthly measurements of nutrients (NOx (nitrate + nitrite), dissolved organic nitrogen (DON), silicate, orthophosphate), chlorophyll *a* (Chl. *a*), salinity, temperature, dissolved oxygen (DO) and Secchi disk depth (as a proxy for water clarity). Nutrient analyses were performed by SJRWMD or its contract laboratories using standard EPA protocols (APHA, 1989). Dissolved organic nitrogen (DON) was calculated by subtracting ammonium from the total dissolved Kjeldahl nitrogen. Rainfall data was obtained from NASA's Kennedy Space Center long term rain gauge data (http://www.trmm.ksc.nasa.gov/). NASA samples 24 'rain mills' located within a 25 miles radius of the southern Mosquito Lagoon and northern Indian River Lagoon. During our analyses, we discovered unusual environmental conditions in 2011 as well as 2012 and thus considered these two years relative to mean values from 1996 to 2010. For all of the archived data including rainfall, monthly means were calculated for 1996–2010 for each parameter which were then compared to monthly observations during 2011 and 2012. Temporal water quality comparisons were performed at a representative station in the southern Mosquito Lagoon where the brown tide was most intense in 2012 (IRLML02; Fig. 1), whereas the spatial dynamics of Chl. *a*, salinity, water clarity (via secchi disk) and dissolved oxygen across the Mosquito and northern Indian River Lagoons were assessed for the summer and fall of 2012. Specifically, point data from sampling stations (Fig. 1) was used to produce spatial distribution maps in ESRI® ArcGIS® 10, using the Geostatistical Analyst extension, and an ordinary Kriging algorithm to interpolate between random point data. To assess the impacts of the brown tide event on the Mosquito Lagoon fish populations, fish kills in Volusia County, FL, as reported by the Florida Fish and Wildlife Conservation Commission were assessed monthly from 1996 to 2012 (http://research.myfwc.com/fishkill/). Impacts of the blooms on resident bivalve populations were assessed by interviewing personnel at local shellfish hatcheries and accessing a USDA Farm Service Agency, Noninsured Disaster Assistance Program report made in the Mosquito Lagoon during the fall of 2012.

2.4. Field samples

During the 2012 sampling, physical parameters (temperature, salinity, and dissolved oxygen) were measured using a Hydrolab Minisonde® Multiprobe SE, and light attenuation was assessed with a Secchi disk at each station. Water was collected in acid-cleaned (10% HCl), 20 L polyethylene carboys that were brought to the Marine Discovery Center in New Smyrna Beach, FL, for processing. Size fractionated Chl. *a* was determined by filtering triplicate samples onto 0.2 and 2 μm polycarbonate filters. These filters were stored frozen until subsequent analysis via standard fluorometric methods (Welschmeyer, 1994). Whole water samples were preserved with 10% buffered formalin and analyzed flow cytometrically to assess picoplankton densities (Olson et al., 1991). Abundances of cyanobacteria and photosynthetic picoeukaryotes were determined using a Fluorescence Activated Cell Scan (FACScan; Becton, Dickinson and Company) flow cytometer using

fluorescence patterns and particle size from side angle light scatter (Olson et al., 1991). *A. lagunensis* abundances were enumerated using the species-specific antibody developed by Lopez-Barreiro et al. (1998) modified for use on a flow cytometer. Cell abundances of plankton groups were normalized to carbon using 200 and 250 fg carbon cell^{-1} for cyanobacteria and picoeukaryotes, respectively (Kana and Glibert, 1987; Liu et al., 1999; Worden et al., 2004) and 900 fg carbon cell^{-1} for *Aureoumbra lagunensis* (Liu et al., 2001).

2.5. Clearance rate experiments with natural bloom water

To assess the impacts of the Florida brown tides on resident shellfish populations, clearance rate experiments were performed according to the 'clearance method' (Shumway et al., 1985; Riisgard, 2001; Harke et al., 2011). Clearance rates were assessed for the northern quahogs (a.k.a. hard clam; *Mercenaria mercenaria*) and eastern oysters (*Crassostrea virginica*) which are native to the Mosquito and Indian River Lagoon systems. Juvenile *M. mercenaria*, were supplied by the Sen. George Kirkpatrick Marine Lab at Cedar Key, FL while *C. virginica* were supplied by Research Aquaculture, Inc. (Ft. Pierce, FL) through the Marine Discovery Center (New Smyrna Beach, FL). Shellfish were acclimated for 48 h under gentle aeration in 0.2 μm filtered seawater (FSW) from the Mosquito Lagoon supplemented with 10^5 *Isochrysis galbana* (Tahitian strain) cells mL^{-1}. Prior to the initiation of experiments, animals were placed in FSW without food under gentle aeration for at least 1 h and then gently transferred to experimental vessels (Harke et al., 2011). For experiments, individuals were placed in 300 mL beakers filled with water from stations IRLML02 and SBL (Fig. 1) collected on 24 and 26 September and FSW from IRLML02 supplemented with *I. galbana* at a biovolume matching 10^6 *Aureoumbra lagunensis* cells mL^{-1} (bio-volumes of *A. lagunensis* and *I. galbana* were 24 and 34 μm^3, respectively). Triplicate control beakers of each experimental water type without animals were established to account for changes in cell densities not due to animal activity. Initial cell concentrations within experimental vessels were assessed by preserving separate 4.5 mL aliquots with 1% formalin (final concentration) and 1% glutaraldehyde (final concentration). Individual *M. mercenaria* (17.1 ± 2.9 mm) were placed within beakers, whereas five *C. virginica* individuals (6.1 ± 2.0 mm length) were placed in each beaker due to their significantly smaller size. *C. virginica* individuals were allowed to feed for 1 h from the time they were placed in the beaker, whereas *M. mercenaria* fed for 1 h from the time each individual animal was observed to open and/or extend their siphon. After the 1 h experiment, samples for the quantification of final cell abundances were preserved in formalin and glutaraldehyde and the animals were frozen, dried at 55 °C, weighed, and combusted at 450 °C and weighed again to determine ash-free, dry tissue weight. The dry tissue weight, initial cell densities, and final cell densities were used to determine a clearance rate for each species in L h^{-1} g^{-1} according to Coughlan (1969).

2.6. Clearance rate experiments with cultures of A. lagunensis

In addition to experiments with brown tide bloom water from the Mosquito Lagoon, clearance rates of cultures of *Isochrysis galbana* and *Aureoumbra lagunensis* (strain Al-FL-01; isolated from the Mosquito Lagoon) were performed largely as described above. Exceptions included the use of larger, juvenile *Mercenaria mercenaria* and *Crassostrea virginica* (21.5 ± 3.6 mm length and 35 ± 4.1 mm length, respectively; obtained from the East Hampton Shellfish Hatchery, NY, USA) at one per vessel. To mimic bloom conditions, cell densities of 4×10^5 and 1×10^6 *A. lagunensis* cells mL^{-1} and a biovolume equivalent of *Isochrysis galbana* at 5×10^5 cells mL^{-1} were used in experiments.

2.7. Monitoring oyster reefs in the Mosquito Lagoon

As part of a long-term, community-based oyster reef restoration project in the northern Canaveral National Seashore (80°51′ W; 28°56′ N) of Mosquito Lagoon that began in 2007, stabilized oyster shells were deployed in intertidal areas in warm months (mid-May through mid-July) and then monitored annually during the cold months (January–March). Recruitment of *Crassostrea virginica* in this system occurs annually from mid-May until temperatures approach an annual minimum (usually December; L. Walters, pers. obs.). In addition to counting the total number of oyster recruits in thirty, randomly selected 0.25 m^2 quadrats on restored reefs at low tide, we recorded the shell lengths in mm of all live oyster recruits in thirty 0.25 m^2 quadrats using callipers.

2.8. Data analysis

Differences between clearance rates were evaluated using one-way analyses of variance (ANOVA) where diet (bloom water, *Aureoumbra lagunensis* culture, *Isochrysis galbana* culture) was the main treatment effect examined and differences between varying diets were examined using Tukey's multiple comparison tests. One-way ANOVAs or Student's *t*-tests were used to compare 2011 and 2012 to long term data (1996–2010). A *p*-value of 0.05 was used to establish significant differences for all tests. Differences in the sizes of oysters recruiting to reefs in the Mosquito Lagoon among years (2009–2012) were compared with a one-way ANOVA followed by Tukey's multiple comparison tests.

3. Results

3.1. Comparison of 2012 to the historical dataset

Precipitation in eastern, central FL has a distinct seasonal pattern. The total annual rainfall in 2011 exceeded the previous 15 year (1996–2010) mean by ~10% (113 cm vs. 100 cm; Fig. 2A). This wetter than average year was followed by a 6-month dry period with significantly ($p < 0.05$) less rainfall (13.0 cm between November 2011 and April 2012) compared to the long term mean (28.2 cm; Fig. 2A). With the exception of July 2012 when precipitation amounts were normal, precipitation from May to August 2012 was slightly higher than the 15 year mean (Fig. 2A). Salinity in the Mosquito Lagoon was up to 50% higher than mean values during 2011 and 2012 ($p < 0.0001$; Fig. 2B). Salinity displayed a seasonal trend with the highest values observed in June 2011 and May 2012 (43.9 and 45.4, respectively) and lower salinities occurring in April and November 2011 and November 2012 (36.6, 38.2 and 37.6, respectively; Fig. 2B). Hypersaline conditions were present throughout the Mosquito and northern Indian River Lagoons during the summers of 2011 and 2012 (Fig. 3). Water temperatures in 2011 were not different from the long term mean. In 2012, temperatures were 33% below average in January, above average from February to May, and near normal for the rest of the year (Fig. 2C).

Large algal blooms occurred within the Mosquito Lagoon in 2011 and 2012 (Figs. 4A and 5). While the mean monthly Chl. *a* levels from 1996 to 2010 ranged from a low of 2.4 μg L^{-1} in February to a high of 7.5 μg L^{-1} in September with little interannual variability, in 2011 and 2012, Chl. *a* levels were more than an order of magnitude higher (Fig. 4A). In 2011, an algal bloom began in August and persisted through December with Chl. *a* levels ranging from 48 to 86 μg L^{-1}, concentrations significantly greater than the long term mean of 5 μg L^{-1} ($p < 0.001$; Fig. 4A). While Chl. *a* levels were near-normal during March–June of 2012, in July, August, and September, a second, significantly more intense phytoplankton bloom developed with monthly Chl. *a* levels of 59,

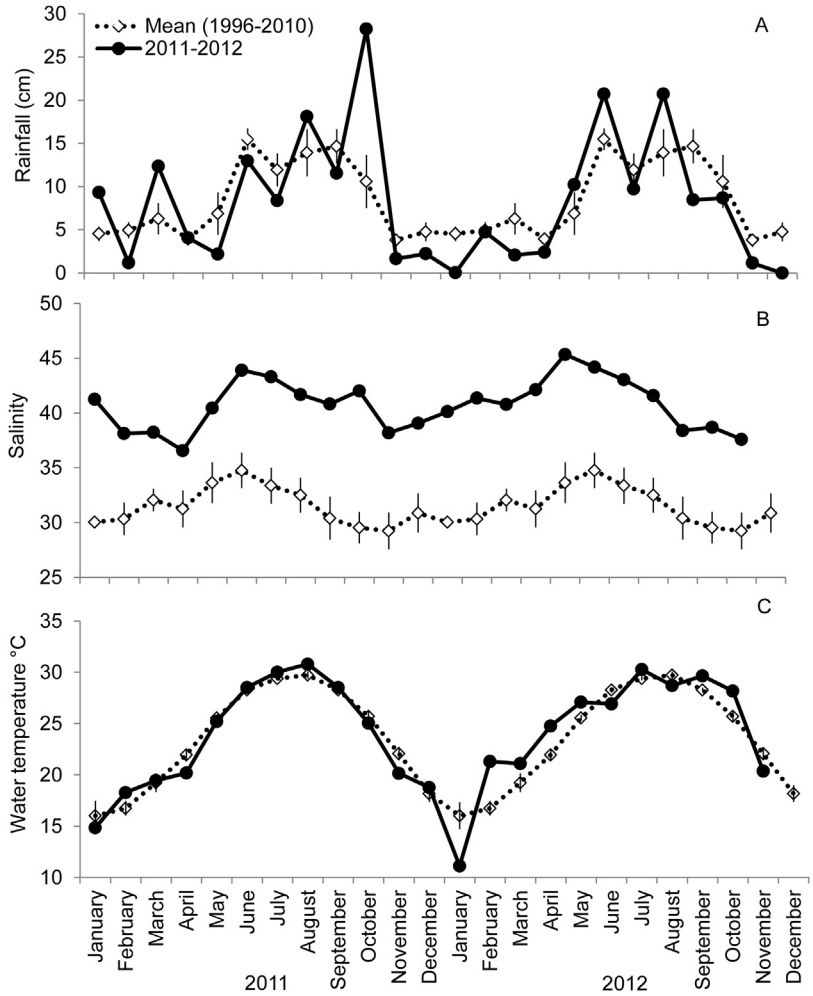

Fig. 2. Physical parameters including (A) rainfall (B) salinity and (C) temperature at station IRLML02 in Mosquito Lagoon, FL (see Fig. 1). Values shown are monthly observations from 2011 to 2012 and monthly means ± standard error from 1996 to 2010. Data was obtained from the St. Johns River Water Management District and NASA weather stations on Cape Canaveral (see Section 2 for details).

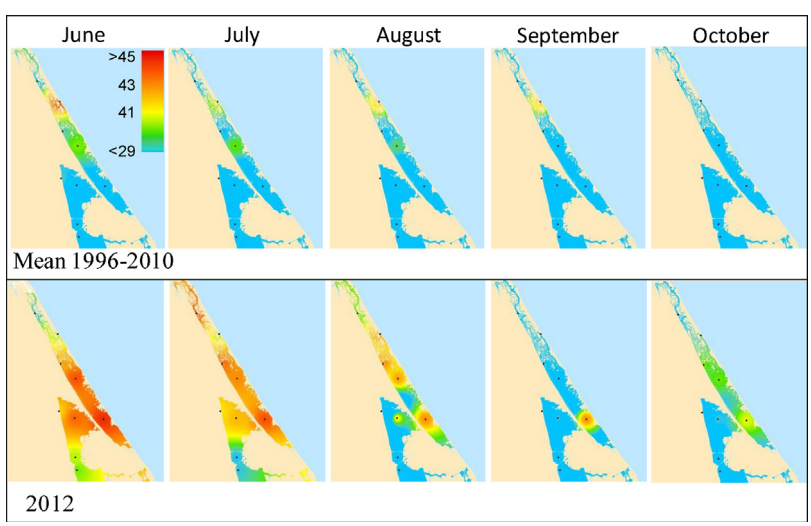

Fig. 3. Salinity distribution in the northern Indian River and Mosquito Lagoon systems. Data was obtained from the St. Johns River Water Management District. The data for 1996–2010 (top panels) is compared to 2012 (bottom panels), the year of the first recorded brown tide in this system. Values depicted on the heat map are salinities and range from <29 (lightest blue) to >45 (darkest red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

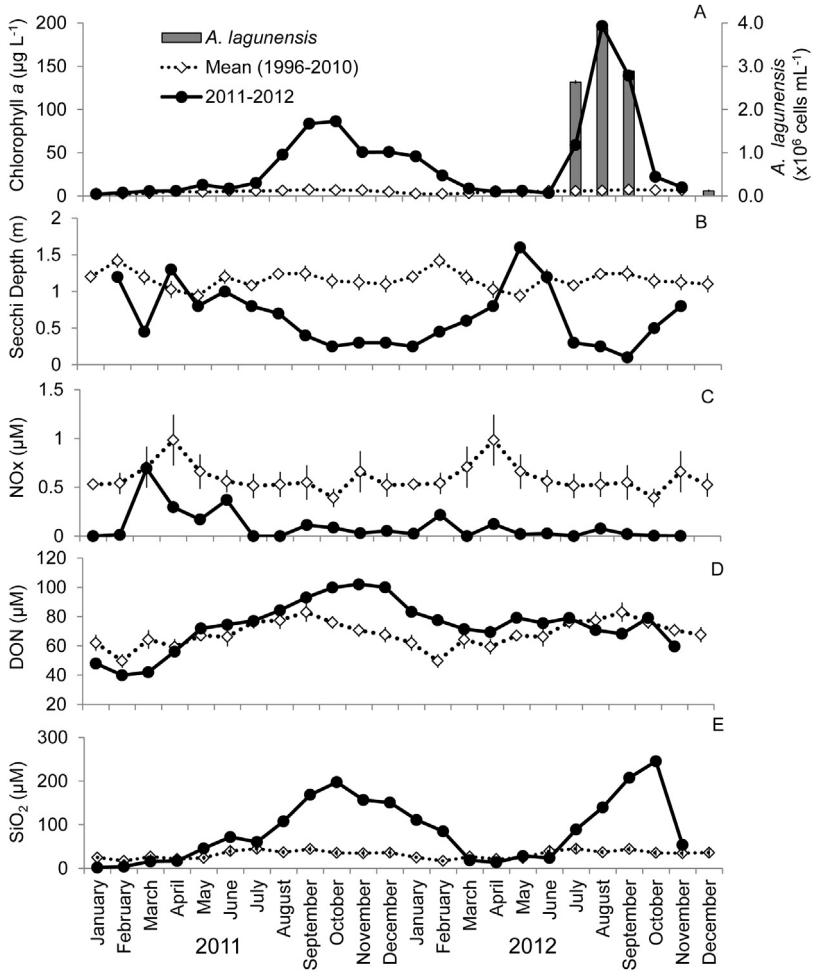

Fig. 4. (A) Concentrations of chlorophyll *a* and *A. lagunensis* abundance, (B) secchi disk depth, (C) nitrate + nitrite (NOx), (D) dissolved organic nitrogen (DON) and (E) silicate (SiO_2) at station IRLML02 compiled as described in Fig. 2.

196, and 139 $\mu g\,L^{-1}$, respectively ($p < 0.001$; Fig. 4A). This unusually dense algal bloom spread throughout the northern Indian River Lagoon and Mosquito Lagoon and began to decline during October (Fig. 5), with Chl. *a* levels remaining at or above 10 $\mu g\,L^{-1}$ through December of 2012 (Fig. 4A). Concurrent with the increases in Chl. *a* from the 2011 and 2012 blooms, water clarity declined dramatically within the Mosquito Lagoon as evidenced by the shoaling of the Secchi disk depth from near

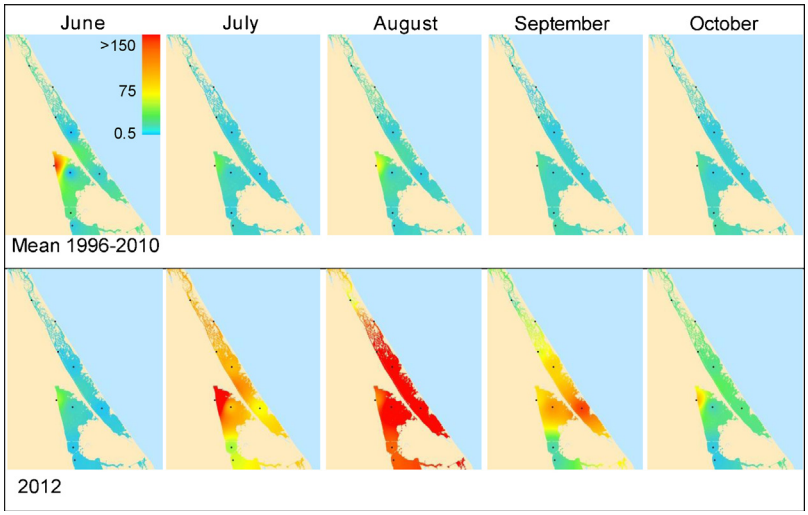

Fig. 5. Chlorophyll *a* distribution in the northern Indian River and Mosquito Lagoon systems. Mean data for 1996–2010 (top panels) are compared to 2012 (bottom panels), the year of the first recorded brown tide bloom. Chlorophyll *a* values ranged from 0.6 (lightest blue) to 300 $\mu g\,L^{-1}$ (darkest red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

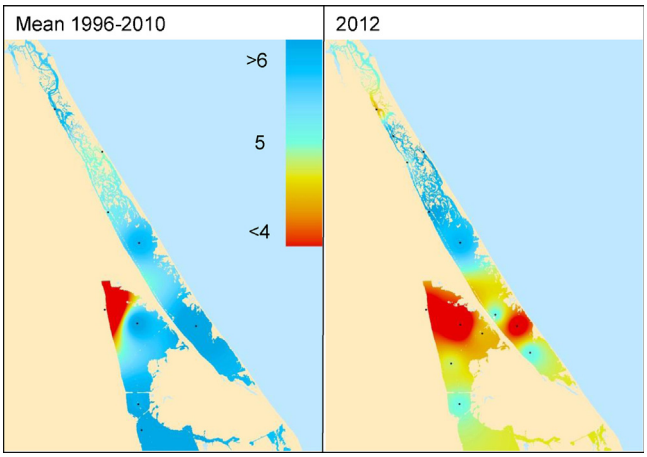

Fig. 6. Dissolved oxygen concentrations in September in the northern Indian River and Mosquito Lagoon systems. Long-term mean September dissolved oxygen levels obtained from the St. Johns River Water Management District from 1996 to 2010 (left) are compared to a 24–26 September 2012 survey (right). Dissolved oxygen concentrations ranged from <4 mg L^{-1} (darkest red) to >6 mg L^{-1} (lightest blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

average values of 0.8 m in May 2011 to 0.3 m in October 2011 during the bloom preceding the brown tide (Fig. 4B). Although Secchi disk depths were near or deeper than average during the spring of 2012, they declined to the shallowest depths on record (0.1 m) in the Mosquito Lagoon during the 2012 brown tide and remained below average through November 2012 (Fig. 4B).

Nitrate concentrations in the Mosquito Lagoon from March 2011 through November 2012 were significantly lower than the 1996–2010 monthly means, with concentrations ranging from undetectable to 0.63 μM ($p < 0.001$; Fig. 4C). From April 2011 through June 2012, DON concentrations were significantly higher than the 1996–2010 monthly means peaking in November 2011 as the 2011 bloom collapsed (100 μM, $p < 0.001$; Fig. 4D). In July 2012, concentrations declined slightly and remained significantly lower than the mean concentration until October 2012, when they recovered to average concentrations (Fig. 4D). From 1996 to 2010, mean monthly silicate concentrations ranged seasonally from a low of 17 μM in February to a high of 45 μM in July. During 2011 and 2012, however, silicate levels increased during summer and fall to levels 5- and 7-fold higher than normal with peak concentrations occurring in October of both years (198 and 245 μM for 2011 and 2012, respectively; Fig. 4E). During the decline of the brown tide bloom in September 2012, low oxygen levels (<4 mg L^{-1}) were recorded at multiple stations throughout the Mosquito and Indian River Lagoon system (Fig. 6) with percent saturation as low as 48% (Table 1). Dissolved oxygen levels within this system in September 2012 were significantly lower than the long term September mean (1996–2010; $p < 0.001$; Fig. 6).

3.2. Morphological and molecular identification of the Florida brown tide organism

Our sequencing of the 18S rRNA gene from isolates from the Mosquito Lagoon, FL, USA (Al-FL-1) and a monospecific bloom population from the Mosquito Lagoon (KC581797 and KC581798) showed 100% identity to GenBank sequences of *Aureoumbra lagunensis* isolates from TX, USA (HQ710573 and HQ710574; Fig. 7). Notable ultrastructural features of cells from culture isolates included large peripheral chloroplasts, a large vacuole space, and presence of a thick (up to 0.5 μm) extra-cellular

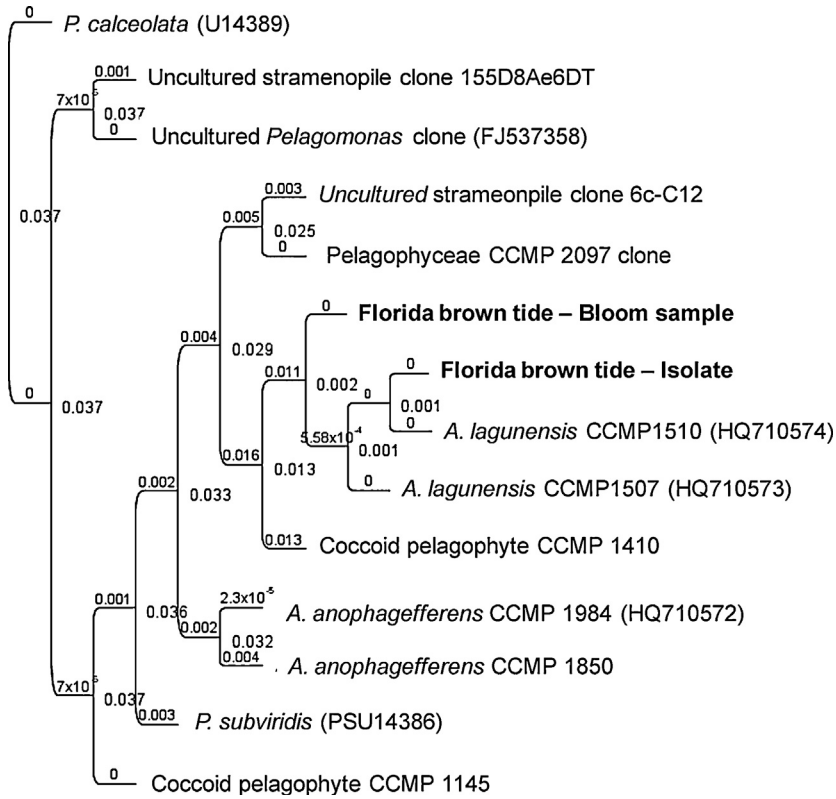

Fig. 7. Phylogenetic, neighbor-joining tree of partial 18s rRNA sequences of the Florida brown tide field sample and culture as compared to other brown tide species and pelagophytes. Florida brown tide culture and field samples were most similar to *Aureoumbra lagunensis* (HQ710574) with distances of 0 and 5.6×10^{-4}, respectively. The field sample had three sequencing ambiguities leading to the slightly greater genetic distance.

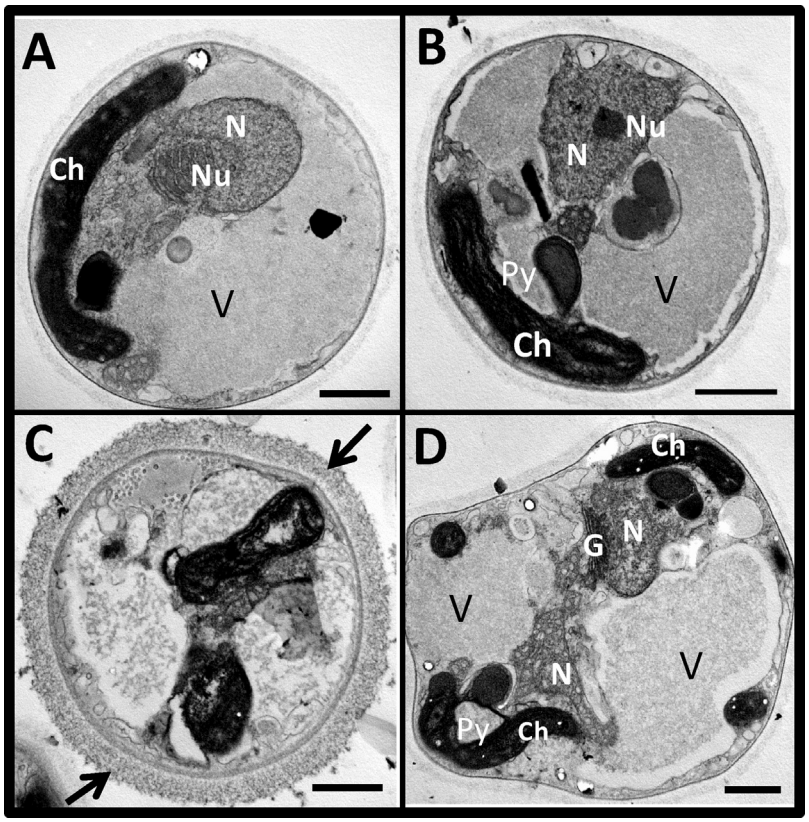

Fig. 8. Transmission electron microscopic micrographs of *Aureoumbra lagunensis* cells of a culture established from a bloom water in the Indian River Lagoon system, Florida. (A and B) Cells with a large chloroplast (Ch) with a pyrenoid (Py), a large vacuole (V), a nucleus (N) and a nucleolus (Nu); (C) a cell with thick extracellular polysaccharides (arrows); (D) a cell in division. Note two separate chloroplasts, two nuclei still connected. Also noted a Golgi body (G), a pyrenoid, and two large vacuoles. Scale bars: 0.5 μm.

polysaccharide coating on cells (Fig. 8), observations consistent with the original description of the TX isolate of *A. lagunensis* (DeYoe et al., 1997). Given this data and the cross reactivity of the bloom-forming organism with the *A. lagunensis*-specific antibody, but not an *A. anophagefferens*-specific anti-body (F. Koch, pers. obs.), we conclude that the brown tide forming organisms in the Mosquito Lagoon and the northern Indian River Lagoon in 2012 were conspecific with the isolates from TX, USA, and thus are *Aureoumbra lagunensis* (Figs. 7 and 8).

3.3. Phytoplankton community composition during the Florida brown tide bloom

During the 2012 brown tide bloom in the Indian River and Mosquito Lagoon systems, *Aureoumbra lagunensis* was the dominant phytoplankter, representing up to 99% of the algal carbon present (Table 2 and Fig. 9). In the southern Mosquito Lagoon, at station IRLML02, cell densities exceeded 2×10^6 cells mL^{-1} from July through September and approached 4×10^6 cells mL^{-1} in August (Fig. 4A). During a spatial survey across the Indian River and Mosquito Lagoon in September, *A. lagunensis* was detected at all stations, ranging from densities of 1×10^4 to 2×10^6 cells mL^{-1} with the highest densities occurring in the southern Mosquito Lagoon stations IRLML02 and SBL (Table 2 and Fig. 9). Concurrent surveys of dissolved oxygen across the Mosquito and northern Indian River Lagoons found undersaturated levels at most locations (3–6 mg L^{-1}; Table 1; Fig. 6). A smaller survey in December 2012 revealed low levels of *A. lagunensis* cells still present throughout the southern Mosquito Lagoon and northern Indian River Lagoon (>10^4 cells mL^{-1}), with peak densities of 10^5 cells mL^{-1} at station IRLML02 (Fig. 4A).

During a spatial survey across the northern Indian River Lagoon and Mosquito Lagoon in September, phytoplankton biomass was lowest at the northern Indian River Lagoon stations (133 μg C L^{-1} and 176 μg C L^{-1} for IRLI06 and CM1, respectively) and highest in the southern Mosquito Lagoon (1181 and 1762 μg C L^{-1}, IRLML02 and SBL, respectively; Fig. 9). While the phytoplankton community

Table 2
Total chlorophyll *a* (Chl *a*) concentrations, the % of the total Chl *a* found in the >2 μm size fraction, and abundance of *A. langunensis* during the Florida brown tide bloom in September 2012. Station locations are depicted in Fig. 1. Stations at which water was collected for clearance rate experiments are marked with an asterisk. BDL denotes bellow detection limit (~1000 cells mL^{-1}).

Station	Date	Chl. *a* (μg L^{-1})	% in >2 μm	*A. lagunensis* (cells mL^{-1})
IRLV05	09/24/12	8.1 ± 0.9	66.1	24,679
GC	09/24/12	12.0 ± 0.6	68.6	12,480
CM65	09/24/12	9.7 ± 0.2	63.8	14,122
IRLV11	09/24/12	15.2 ± 1.5	80.1	40,092
IRLV17	09/24/12	10.8 ± 0.8	76.3	25,008
IRLML169	09/24/12	9.6 ± 1.0	50.0	72,982
CM43	09/24/12	9.7 ± 0.2	80.3	247,566
IRLML02*	09/24/12	38.2 ± 3.9	84.4	1,410,068
SBL*	09/24/12	41.1 ± 1.8	88.5	1,954,019
IRLBFFR	09/25/12	18.2 ± 1.1	61.8	149,060
IRLI02	09/25/12	3.9 ± 0.2	80.4	22,990
CM1	09/25/12	9.7 ± 1.5	74.7	185,399
27010875	09/25/12	3.2 ± 0.1	57.4	BDL
IRLI06	09/25/12	2.5 ± 0.1	55.7	BDL
IRLML169	09/26/12	9.5 ± 1.4	50.0	51,610
CM43	09/26/12	9.1 ± 0.4	80.3	204,213
IRLML02*	09/26/12	35.1 ± 4.9	84.4	1,311,023
SBL	09/26/12	30.8 ± 4.5	88.5	1,390,902

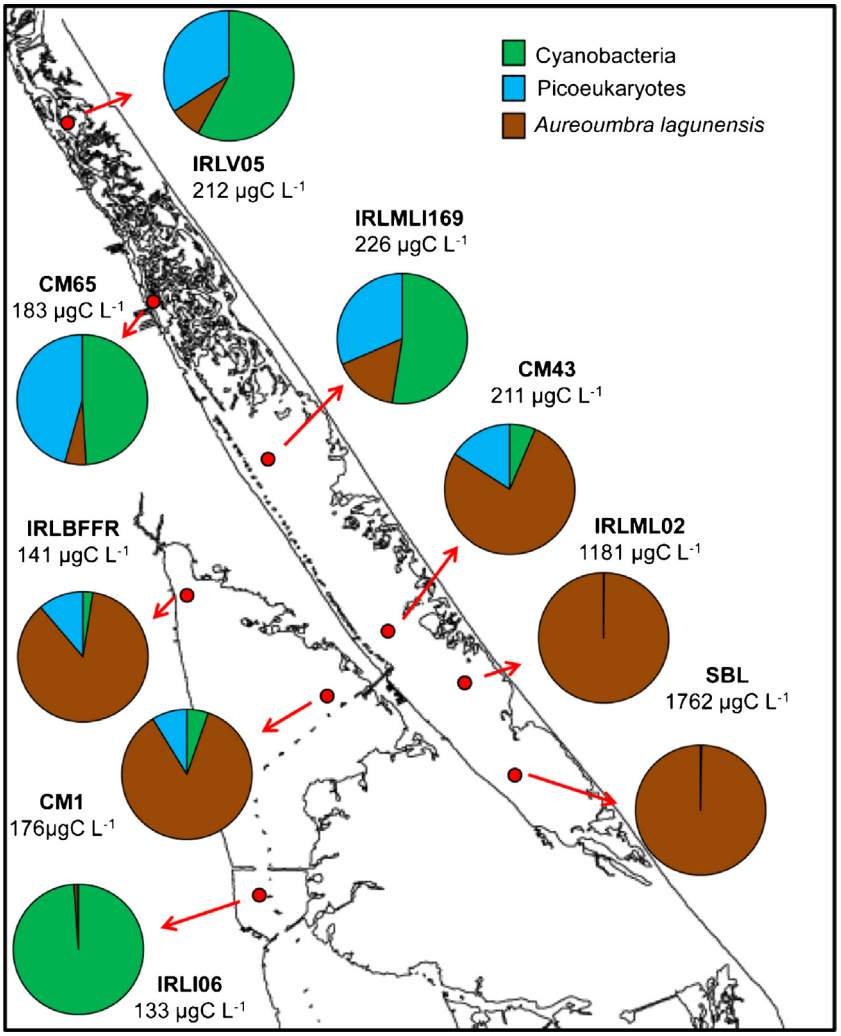

Fig. 9. Carbon-based relative abundance of the major groups of small phytoplankton present in the northern Indian River Lagoon and the Mosquito Lagoon systems during September 2012. The red circles denote the station locations while the brown, green, and blue pie pieces represent the relative abundance of *Aureoumbra lagunensis*, picocyanobacteria, and non-brown tide picoeukaryotic phytoplankton, respectively. Values depicted below each station name give the total carbon concentration in $\mu g\ L^{-1}$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

in the northern Mosquito Lagoon (stations IRLV05, IRLMLI169, and CM65) was co-dominated by cyanobacteria and small picoeukaryotes, it was comprised of more than 99% of *Aureoumbra lagunensis* carbon at southern stations, IRLML02 and SBL (Fig. 9). In contrast to the Mosquito Lagoon, the northern Indian River Lagoon phytoplankton community (i.e. IRLI06) was dominated (>99%) by cyanobacteria (Fig. 9). Stations in the southern Mosquito Lagoon dominated by *A. lagunensis* generally had a larger fraction of Chl. $a > 2\ \mu m$ (>85%) than stations with low levels of *A. lagunensis* to the north and south (<75%; Table 2).

3.4. Clearance rates of bivalves in the presence of brown tide bloom water

Clearance rates of *Mercenaria mercenaria* and *Crassostrea virginica* were significantly ($p < 0.001$ for both) lower in the presence of Mosquito Lagoon water with bloom densities of *Aureoumbra lagunensis* compared to *Isochrysis galbana*. For example, clearance rates of *M. mercenaria* fed a monoculture of *I. galbana* were $1.86 \pm 0.97\ L\ h^{-1}\ g^{-1}$, but were more than an order of magnitude lower when fed bloom water from the Mosquito Lagoon

Table 3
Clearance rates (measured in liters per gram dry weight per hour) of northern quahogs (=hard clam, *M. mercenaria*) and eastern oyster (*C. virginica*) exposed to Mosquito Lagoon water was collected on 24 September 2012 from station SBL (*A. lagunensis* = 1.9×10^6 cells mL^{-1}) and IRLML02 on 24 and 26 September, 2012 (*A. lagunensis* = 1.4×10^6 cells mL^{-1} on both sampling dates). The *I. galbana* treatment reflects clearance rates when the bivalves were exposed to *I. galbana* (7×10^5 cells mL^{-1}) $_{-1}$) in filtered Mosquito Lagoon seawater, while ND denotes 'no data'. Asterisks indicate rates significantly lower than the *I. galbana* control ($p < 0.05$). Values are mean ± standard deviation ($n = 6$).

	I. galbana 7×10^5 (cells mL^{-1})	SBL 9/24 1.9×10^6 (cells mL^{-1})	IRLML02 9/24 1.4×10^6 (cells mL^{-1})	IRLML02 9/26 1.4×10^6 (cells mL^{-1})
M. mercenaria	1.86 ± 0.97	0.20 ± 0.33*	0.20 ± 0.48*	ND
C. virginica	2.19 ± 1.09	0.01 ± 0.09*	0.12 ± 0.29*	0.05 ± 0.12*

Table 4
Clearance rates (measured in liters per gram dry weight per hour) of two species of bivalves, the hard clam (*M. mercenaria*) and eastern oyster (*C. virginica*) exposed to cultures of *A. lagunensis* (Strain Al-FL-01, isolated from IRLML02) and *I. galbana*. Concentrations of *I. galbana* were 5×10^5 cells mL^{-1} (= to *A. lagunensis* biovolume of 9×10^5 cells mL^{-1}) while '*A. lagunensis*, low' and '*A. lagunensis*, high' contained 4×10^5 and 1×10^6 cells mL^{-1} of *Aureoumbra*, respectively. Stars indicate rates significantly lower than the *I. galbana* control ($p < 0.05$). Values are mean ± standard deviation ($n = 6$).

	I. galbana 5×10^5 (cells mL^{-1})	*A. lagunensis*, low 4×10^5 (cells mL^{-1})	*A. lagunensis*, high 1×10^6 (cells mL^{-1})
M. mercenaria	0.50 ± 0.13	0.04 ± 0.09*	0.01 ± 0.03*
C. virginica	1.27 ± 0.46	0.18 ± 0.07*	0.04 ± 0.03*

(0.20 ± 0.33 and 0.20 ± 0.47 L h^{-1} g^{-1}, from SBL (9/24) and IRLML02 (9/24), respectively; $p < 0.001$; Table 3). Similarly, clearance rates of *C. virginica* fed a monoculture of *I. galbana* were high (2.19 ± 1.09 L h^{-1} g^{-1}) but were significantly lower when fed bloom water from the Mosquito Lagoon (<0.15 L h^{-1} g^{-1}; $p < 0.001$; Table 3).

3.5. Clearance rates of bivalves fed cultures of A. lagunensis

Consistent with experiments using natural bloom water, bivalves fed a monoculture of *Aureoumbra lagunensis* isolated from the Mosquito Lagoon displayed significantly lower clearance rates compared to individuals fed *Isochrysis galbana* ($p < 0.001$ for all; Table 4). Clearance rates of *Mercenaria mercenaria* fed *I. galbana* were 0.50 ± 0.13 L h^{-1} g^{-1} compared to 0.04 ± 0.09 and 0.01 ± 0.03 L h^{-1} g^{-1} when fed low (4×10^5 cells mL^{-1}) and high (1×10^6 cells mL^{-1}) concentrations of *A. lagunensis* (Table 4). Similarly, *Crassostrea virginica* displayed significantly lower clearance rates when exposed to both low and high concentrations of *A. lagunensis* (0.18 ± 0.07 and 0.04 ± 0.03 L h^{-1} g^{-1}, respectively; $p < 0.001$ for both; Table 4) when compared to a diet of *I. galbana* (1.27 ± 0.41 L h^{-1} g^{-1}).

3.6. Impacts of the Florida brown tide within the Mosquito Lagoon

The brown tide that occurred in the Mosquito Lagoon during the summer of 2012 negatively impacted resident finfish and shellfish populations. The mean recruitment density of *Crassostrea virginica* on intertidal reefs in the Mosquito Lagoon in 2012 was 79 ± 8.0 oyster recruits m^{-2}; this was lower, but not significantly different, than 2007–2011 (mean = 140 ± 54 recruits m^{-2}; range = 18–330 recruits m^{-2}; L. Walters, unpublished data). In contrast, the size of the newly settled *C. virginica* on reefs in 2012 was significantly smaller (32 ± 1.7 mm) compared to 2009, 2010 and 2011 (all >45 mm; $p < 0.001$; Tukey test; Fig. 10). In addition, multiple shellfish aquaculture facilities in Mosquito Lagoon reported mass mortalities of cultured and outplanted shellfish. For example, Cedar Creek Shellfish Farms (New Smyrna Beach, FL) had planted hard clams (*Mercenaria mercenaria*) surveyed before and after the brown tide event by the USDA, Farm Service Agency, Non-Insured Disaster Assistance Program (NAP). This program reported a loss of 5 million hard clams and a 91% mortality rate during the August and September brown tide for Cedar Creek Shellfish Farms. Finally, prior to 2012, fish kills were infrequently reported within the Mosquito Lagoon in Volusia County, FL, with an average of one report per month during summer (June–September; Table 5). During the summer of 2012, there were eight fish kills reported in this region in July and more than 30 reported in September (Table 5) as the brown tide waned.

Table 5
Mean monthly number of fish kills reported 1996–2011 in Volusia County, FL, by the Florida Fish and Wildlife Conservation Commission compared to reports in 2012. Months during which the brown tide bloomed are bold.

	Mean (1996–2011)	2012
JAN	3	0
FEB	1	0
MAR	1	0
APR	1	1
MAY	0	3
JUN	1	2
JUL	1	**8**
AUG	1	**1**
SEP	2	**31**
OCT	2	1
NOV	1	1
DEC	1	0

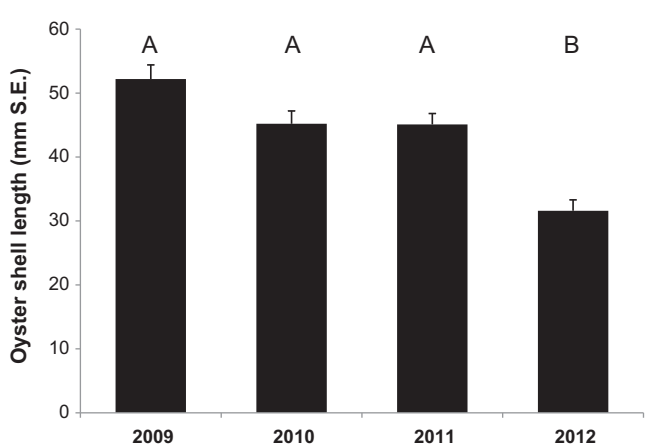

Fig. 10. Shell length (mean ± S.E.) of newly settled eastern oysters (*C. virginica*) on oyster reefs in the Mosquito Lagoon. Letters represent years that are not different from (A) and significantly different from (B) other years ($p < 0.05$; Tukey test).

4. Discussion

While brown tides of the pelagophyte *Aureoumbra lagunensis* have occurred in TX, USA, for more than 20 years, the blooms described in this study represent the first blooms outside of this state. While these blooms were unprecedented within the Mosquito and the northern Indian River Lagoons, they shared many similarities with bloom events that have occurred in TX. At the same time, there were several attributes of the brown tide in FL that have never been reported for any pelagophyte-caused brown tide. This report, therefore, sheds new light on the phenomenon of brown tides caused by pelagophytes including *Aureoumbra lagunensis* as well as the potential for these events to reoccur in Florida estuaries and elsewhere in the southeast US.

While initial observations of the brown tide in the Mosquito and Indian River Lagoons suggested the bloom might be caused by a pelagophyte (E. Philips. U. Florida; pers. comm.), sequencing of the 18S rRNA gene provided unambiguous evidence that the bloom forming organism was the pelagophyte, *Aureoumbra lagunensis*. Beyond this genetic evidence, antibodies developed specifically for

A. lagunensis (Lopez-Barreiro et al., 1998) cross-reacted with cells from the Mosquito Lagoon, while the anti-bodies developed for another pelagophyte, *Aureococcus anophagefferens* (Caron et al., 2004), did not. Multiple attributes of the cells observed from the blooms and the monocultures established from the Mosquito Lagoon including the cell diameter (4–5 μm), the absence of a flagella, and its extended vacuolar space are more consistent with *A. lagunensis* than other pelagophytes such as *A. anophagefferens*, *Pelagomonas calceolata*, and *Pelagococcus subviridis* (Gobler and Sunda, 2012; Fig. 8C).

4.1. Environmental conditions associated with the Aureoumbra lagunensis bloom

Historically, the Mosquito Lagoon has been an oligotrophic estuary, characterized by low levels of Chl. *a* and a clear water column (Fig. 4; Phlips et al., 2002; Badylak and Phlips, 2004). Multiple attributes of this system changed radically during the 2011 and 2012 and these changes may have favored the formation of the brown tide caused by *A. lagunensis*. While salinities in this system have typically ranged from 30 to 34 over an annual cycle, this estuary became hypersaline for much of 2011–2012 (Figs. 2B and 3). This condition was partly established in 2010, a dry year, when salinities increased to >40 by December of that year. Hypersalinty, however, was most severe prior to the initiation of the brown tide when rainfall levels were substantially below average, temperatures were above average (November 2011 through May 2012) and salinities rose from 38 to 45 (Figs. 2 and 3). Low rainfall during this period was likely to have a larger impact on estuarine salinity than summer rain events due to the low rates of evapotranspiration during winter and early spring. Hypersalinity established during the spring was likely maintained by the relatively low tidal exchange with the ocean (Sheng et al., 1990; Smith, 1993) and high evapotranspiration rates during summer. Regardless, *A. lagunensis* grows at near maximal rates over a very wide range of salinities (10–90) and the initial *A. lagunensis* bloom in the Laguna Madre estuary (TX) occurred after a prolonged dry period and under the hypersaline conditions (salinities >40; Buskey et al., 1998). There was a significant negative correlation between microzooplankton abundance and salinity in the Laguna Madre during brown tides there and many phytoplankton grow poorly under hypersaline conditions (Buskey et al., 1998). As such, the *A. lagunensis* bloom in the Mosquito Lagoon may have been partly related to hypersalinity that produced low growth rates of competing algae and low microzooplankton grazing pressure on *A. lagunensis*.

The nutrient regime present in the Mosquito Lagoon during 2012 was likely to have favored rapid growth for *Aureoumbra lagunensis* but perhaps not for co-occurring phytoplankton. Nitrate levels were substantially lower than average during the bloom likely due, in part, to lower input rates associated with reduced rainfall during early 2012. *A. lagunensis* is capable of growing maximally at very low concentrations of inorganic nutrients (Liu et al., 2001) that are limiting to other small phytoplankton (Sunda and Hardison, 2007, 2010) but is unable to grow on nitrate (DeYoe and Suttle, 1994). Hence, the drop in nitrate concentrations to undetectable levels was likely of little consequence for *A. lagunensis* but would have negatively impacted the growth of competing phytoplankton, perhaps opening a niche for *A. lagunensis* to fill. Dissolved organic nitrogen (DON) levels in the Mosquito Lagoon rose to extremely high levels prior to the onset of the *A. lagunensis* bloom, potentially associated with the remineralization of the large phytoplankton bloom that collapsed in late 2011 and early 2012. Given the ability of *A. lagunensis* to grow rapidly on organic nitrogen and phosphorus substrates (Muhlstein and Villareal, 2007), the higher than average DON levels in the Mosquito Lagoon may have given it a nutritional advantage over other, strictly autotrophic algae. Interestingly, while DON levels were above average during the initiation of the *Aureoumbra* bloom, levels declined during late summer when *A. lagunensis* densities were maximal. This pattern is counter to the dynamics of DON observed in both the long term monthly averages and the trend in 2011 when levels steadily increased through the summer. Given that reductions in DON occurring during the peak of brown tides caused by another pelagophyte, *A. anophagefferens*, have been interpreted as utilization of DON compounds by the bloom (LaRoche et al., 1997; Gobler et al., 2004a,b, 2011), it is possible that declining DON levels in the Mosquito Lagoon during August and September of 2012 were indicative of the organic nitrogen use by *A. lagunensis*. The significantly higher than average silicate levels during the summers of 2011 and 2012 in the Mosquito Lagoon were likely reflective of the absence of diatoms during these blooms, as the 2011 algal bloom was caused by pedinophytes (SJRWMD 2011) and diatoms were absent during the summer and early fall of 2012. High silicate levels were also likely due to enhanced benthic fluxes of silicate potentially associated with the greater than usual organic substrate from blooms settling on the benthos leading to increased microbial remineralization (Cowan and Boynton, 1996).

Other environmental factors that may have favored the dominance of *Aureoumbra lagunensis* in the Mosquito and Indian River Lagoons during the summer of 2012 may have included light and temperature. Light levels during the brown tide were the lowest on record in the Mosquito Lagoon, but *A. lagunensis* maintains near-maximum growth rates at a light intensity (50 μmol photons m^{-2} s^{-1}; 2% of noon solar irradiance; Sunda and Hardison, 2010) that is limiting to many other phytoplankton (MacIntyre et al., 2004; Gobler et al., 2011). Hence, the extremely low light levels caused by the onset of the brown tide bloom in the Mosquito and Indian River Lagoons during the summer of 2012 may have favored the further proliferation of *A. lagunensis* over other phytoplankton that require higher light levels. Furthermore, while many phytoplankton reach their maximal growth rates at 20–25 °C (Eppley, 1972; Behrenfeld and Falkowski, 1997), the specific growth rates of *Aureoumbra* increase with temperature from 0.25 d^{-1} at 15 °C to 0.62 d^{-1} at 25 °C and 0.87 d^{-1} at 30 °C (Gobler and Sunda, 2012). Thus, the highest summer temperatures in the Mosquito and Indian River Lagoons favored the rapid growth of *Aureoumbra*, but perhaps not some other algae. Additionally, the declines in temperature that occurred during the fall may have permitted other algal species, better adapted to lower temperatures, to succeed *A. lagunensis*, an occurrence that differed from the 8-year continuous bloom within Laguna Madre from 1990 to 1997 (Buskey et al., 1997). The succession of *A. lagunensis* by picocyanobacteria in FL as evidenced by the presence of high cyanobacterial densities during late September in regions which had dense brown tide during August parallels observations of another brown tide forming pelagophyte, *A. anophagefferens*, which is typically succeeded by *Synechococcus* sp. in NY estuaries (Gobler et al., 2004a,b, 2011).

4.2. Ecological impacts of the Aureoumbra lagunensis bloom

The ecological impacts of *Aureoumbra lagunensis* blooms in TX, USA, have included a decreased abundance of seagrass beds (Onuf, 1996), a decrease in the biomass and diversity of benthic invertebrates (Ward et al., 2000), and the collapse of populations of the dominant filter feeder, the surf clam, *Mulinia lateralis* (Montagna et al., 1993). Consistent with the latter finding, we found that juveniles of the primary filter feeding bivalves within the Mosquito Lagoon, *Mercenaria mercenaria* and *Crassostrea virginica*, cleared brown tide bloom water and isolates of the

Florida *A. lagunensis* at rates significantly lower than those displayed by individuals fed *Isochrysis galbana* at identical total biovolumes. Similar impacts on bivalves by another brown tide-forming pelagophyte, *A. anophagefferens*, in NY, USA, have led to the collapse of the *A. irradians* fishery and mass mortality of *M. mercenaria* populations (Bricelj and Kuenstner, 1989; Gobler et al., 2005). The secretion of extra-cellular polymer substances by brown tides as observed in *A. lagunensis* isolated from FL (Fig. 8) has been implicated in the inhibition of feeding by grazers (Buskey et al., 1997; Bricelj et al., 2001). During the 2012 brown tide in the Mosquito Lagoon, wild and cultured populations of *M. mercenaria* and *C. virginica* experienced significant mortalities (USDA 2012; Mike Sullivan, Cedar Creek Shellfish Farms, pers. comm.; David Preston, Port Saint John Oyster Harvests, pers. comm.), findings consistent with the results of our clearance rate experiments. Such a loss of benthic filtration pressure within shallow lagoons could indirectly promote the proliferation of algal blooms through positive feedback (Sunda et al., 2006). *C. virginica* that settled on oyster reefs in the Mosquito Lagoon in 2012 were significantly smaller than those that settled from 2009 to 2011. While the newly settled oysters may have grown slower during the brown tide event due to a lack of feeding, given the small variance in sizes among individuals, it seems likely that their settlement occurred late in the season (October through December) once *A. lagunensis* densities had declined significantly. Interestingly, *C. virginica* that settled on oyster reefs in the Mosquito Lagoon in 2011 were not different in size from prior years suggesting that the intense algal bloom in that year was not harmful to eastern oysters. In contrast to sizes, the survival rates of oysters on these reefs in Canaveral National Seashore in 2012 did not differ from rates observed from 2008 to 2011 (L. Walters, pers. obs). It is possible that oysters on these reefs fared better than clams and oysters which persisted in or on sediments where oxygen levels were likely to have been significantly lower compared to the water column. Moreover, recent research has demonstrated that early life stage *C. virginica* are more resistant to brown tides caused by *A. anophagefferens* than *M. mercenaria* (Talmage and Gobler, 2012), suggesting eastern oysters may be more likely to survive brown tides than hard clams.

An unexpected occurrence during the *Aureoumbra lagunensis* brown tide in the Mosquito Lagoon was the widespread occurrence of fish kills as the bloom collapsed in September 2012. Historically, pelagophyte brown tides in TX and NY have not been associated with fish kills and there are no known ichthyotoxic properties associated with *A. lagunensis* cells. Indeed, we have found that, with modest aeration, juvenile sheepshead minnows (*Cyprinodon variegatus*) can survive several weeks of exposure to *A. lagunensis* cultures at bloom densities ($>10^6$ cells mL^{-1};data not shown). During mid-day surveys across the Mosquito Lagoon during late September of 2012, dissolved oxygen levels were found to be uniformly undersaturated and as low as 3 mg L^{-1} (Table 1 and Fig. 6), a level known to cause mortality in some fish species (Vaquer-Sunyer and Duarte, 2008). The severe shading that occurs during brown tides (Buskey et al., 1997), the high levels of labile dissolved organic matter present during blooms in the Mosquito Lagoon, and the possible metabolism of organic matter by *A. lagunensis* during the bloom are collectively likely to minimize photosynthesis and maximize respiration, and thus contribute toward net heterotrophy and low levels of oxygen. Moreover, nighttime levels of dissolved oxygen were almost certainly lower than the values found during mid-day surveys, and were perhaps low enough to contribute toward the 32 reported fish kills that occurred as the brown tide in the Mosquito Lagoon collapsed in September 2012. Finally, there was a large and significant loss of seagrass in the northern Indian River and Mosquito Lagoons during 2011 and 2012 (E. Phlips. U. Florida; Pers. comm.), which may be related to the severe shading associated with the brown tide as has been observed during blooms of *A. anophagefferens* in New York estuaries (Cosper et al., 1987).

4.3. Conclusions

During the past two decades, HABs and brown tides have expanded within the US and across the globe (Hallegraeff, 1993; Trice et al., 2004; Probyn et al., 2001; Zhang et al., 2012). The brown tide in the northern Indian River and Mosquito Lagoons during 2012 represented the first HAB caused by *Aureoumbra lagunensis* outside of TX and the first report of *Aureoumbra* cells on the US east coast. Given the interconnected nature of lagoons along the entire east coast of Florida and Georgia, the further expansion of these blooms to the north and south of the Mosquito and Indian River Lagoons seems plausible. Temperatures globally and on the US east coast are expected to increase in the coming decades (I.P.C.C, 2007). Given that *Aureoumbra* grows maximally at temperatures exceeding 30 °C and its dominance during the warmest months of the year in the Mosquito Lagoon, such continued warming may contribute to the intensification of *A. lagunensis* blooms on the US east coast as has been predicted for other HABs in other regions (Hallegraeff, 2010).

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